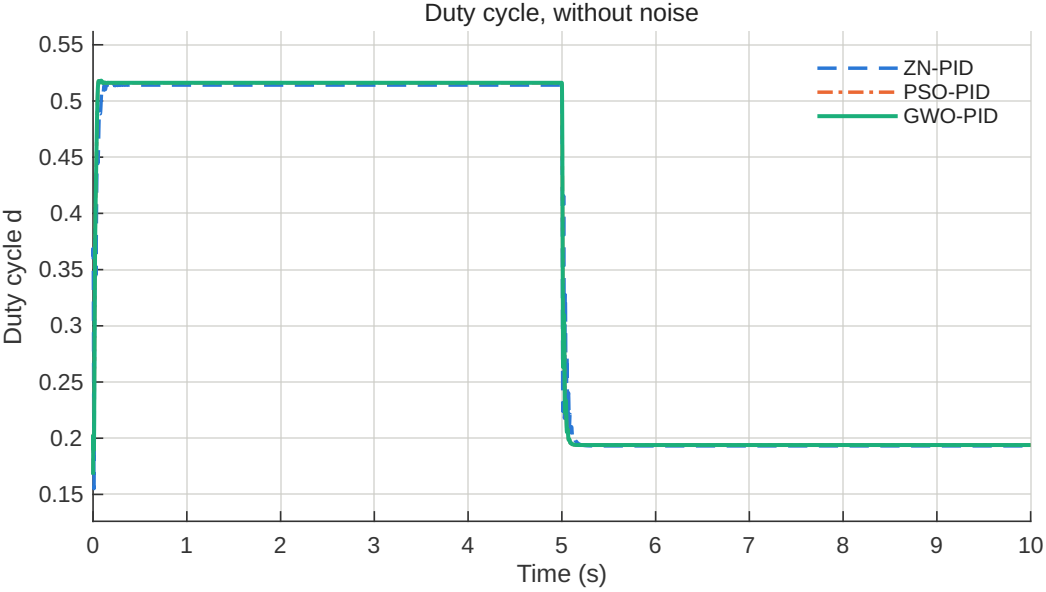

Duty cycle, without noise
ZN-PID
PSO-PID
GWO-PID
Duty cycle d
Time (s)
0.55
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0
1
2
3
4
5
6
7
8
9
10